

The first coupled controlled Boussinesq stage with all physical scales retained

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1 The actual first controlled stage in the original units

The source for this section is Alpöge–Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, <https://cims.nyu.edu/~tristanb/boussinesq.pdf>, released PDF, 76 pages. The source locations used here are Lemma 1.2 and (1.6), pp. 6–7; (3.1)–(3.12), pp. 8–10; the complete proofs of Lemmas 3.7–3.8 and (3.24)–(3.34), pp. 15–18; (3.58) and the following second-stage calculation, p. 27; and the first holding integral on p. 36. The proofs below are given in full. The source’s unit-scale convention is not imposed on the fields in this section.

1.1 Parameters, coordinates, and the original coupled equations

Let $x = (x_1, x_2) \in \mathbb{R}^2$, $e_2 = (0, 1)$, and

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_\alpha = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, \quad e(s) = (\sin s, \cos s).$$

In particular $R_\alpha e(s) = e(s - \alpha)$ and $\operatorname{curl} u = \partial_1 u_2 - \partial_2 u_1$. Retain $A_0 > 0$, $\lambda_0 \geq 1$, and the source’s two scaling parameters

$$\mu = \lambda_0, \quad \nu = \sqrt{A_0}, \quad K_1 = \mu \lambda_1.$$

Here ν is the source’s *time-scaling parameter*. Physical momentum viscosity below is denoted $\nu_{\text{phys}} \geq 0$; thermal diffusivity is $\kappa_{\text{phys}} \geq 0$. The first pulse compression is the source’s Λ_1 , which is different from the physical spatial frequency K_1 .

Fix a smooth nondecreasing step $\eta : \mathbb{R} \rightarrow [0, 1]$ equal to zero on $(-\infty, 0]$ and one on $[1, \infty)$, and $h \in C_c^\infty((0, 1))$ with $h \geq 0$ and $\int_0^1 h = 1$. Write $H = \|h\|_\infty$. The source’s choice is $h = \eta_2$, $c_p = 3$; the following calculation retains any

$$c_p > 1, \quad \Lambda_1 \geq 2c_p H, \quad 0 < s = s_* \leq \frac{1}{8}, \quad c_p \Lambda_1 H \sin s \leq \frac{1}{4}. \quad (1)$$

Since $\int h = 1$, $H \geq 1$ and $\Lambda_1 > 2$. Let $k_1 \geq 0$ be an integer, $\lambda_1 > 1$, and set

$$S = \frac{A_0}{\mu} \lambda_1^{-k_1-6}, \quad L = (k_1 + 5 + \tfrac{1}{8}) \log \lambda_1 \geq 1, \quad P_* = S e^L = \frac{A_0}{\mu} \lambda_1^{-7/8}, \quad \Gamma = \nu \sin s, \quad t_0 = \nu^{-1}. \quad (2)$$

The condition $L \geq 1$ is ensured explicitly by $\lambda_1 \geq \exp((k_1 + 41/8)^{-1})$.

The first layer’s complete preceding affine background, phase transport, and signed amplitude pair are

$$D_{<1} = \dot{\alpha} J, \quad G_{<1} = -A_0 R_\alpha e_2, \quad \dot{M}_1 = D_{<1} M_1, \quad M_1(t_0) = I, \quad (3)$$

$$\zeta_1 = M_1^{-T} e(s), \quad \dot{\Theta}_1 = a_1 \Omega_1, \quad \dot{\Omega}_1 = b_1 \Theta_1, \quad a_1 = -\frac{J\zeta_1 \cdot G_{<1}}{K_1 |\zeta_1|^2}, \quad b_1 = K_1 \zeta_{1,1}. \quad (4)$$

The entry data are

$$\alpha(t_0) = 0, \quad \Theta_1(t_0) = -S, \quad \Omega_1(t_0) = -\frac{K_1}{\nu} S, \quad w_1(t) = \eta(\Gamma(t - t_0)).$$

The activation factor multiplies the physical increment, not the homogeneous amplitude equation. It is flat at t_0 and equals one by $t_0 + \Gamma^{-1}$.

For this actual background, $M_1 = R_\alpha$: differentiation gives $\dot{R}_\alpha = \dot{\alpha} J R_\alpha$ and the initial values agree. Uniqueness follows, for example, by differentiating $R_\alpha^{-1} M_1$. Consequently

$$\zeta_1 = R_\alpha e(s), \quad |\zeta_1| = 1, \quad J\zeta_1 \cdot G_{<1} = -A_0 \sin s, \quad a_1 = \frac{A_0 \sin s}{K_1}, \quad b_1 = K_1 \sin(s - \alpha). \quad (5)$$

The determinant identity used here is explicit: $Je(s) = (-\cos s, \sin s)$ has scalar product $\sin s$ with e_2 , and simultaneous rotation preserves that scalar product. In particular a_1 is constant because both the parent gradient and the new phase rotate; it has not been frozen independently of the evolving parent.

1.2 Exact growth and transition

For the first stage the feedback sums F_0 and F_1 in source (3.4) are empty and equal zero. Thus both the source's growth control $\dot{\alpha} = F_0$ and its transition control $(1 - \eta_b)F_0 + \eta_b F_1$ give $\alpha = 0$. Set

$$t_a = t_0 + \frac{L}{\Gamma}, \quad t_1 = t_a + \Gamma^{-1}, \quad t_b = t_1 + \Gamma^{-1}(1 + \Lambda_1^{-1}).$$

On $[t_0, t_1]$ the exact solution of the original pair is

$$P_1(t) := -\Theta_1(t) = S e^{\Gamma(t-t_0)}, \quad \Omega_1(t) = -\frac{K_1}{\nu} P_1(t). \quad (6)$$

Indeed $a_1 K_1 / \nu = \Gamma$ and $\Gamma K_1 / \nu = K_1 \sin s = b_1$. This proves both differential equations and both initial data. Uniqueness follows from the linear integral equation: on a bounded interval with coefficient norm at most B , the n -fold iterated integral of the difference of two solutions is bounded by its supremum times $(B\ell)^n / n!$, which tends to zero. The same convergent integral series supplies existence on every finite interval. Since P_1 is strictly increasing, t_a is exactly the first crossing of P_* , and the transition contributes exactly one to $\log P_1$. The steering entry amplitude is therefore $P_{\text{ent}} = eP_*$.

1.3 The trial steering family and proof of its unique return

Use $m \in [0, c_p]$ for the pulse trial parameter, to distinguish it from the unchanged spatial parameter $\mu = \lambda_0$. With $\tau = \Gamma(t - t_1)$ and $\tau_b = 1 + \Lambda_1^{-1}$ set

$$z(\tau; m) = \begin{cases} 1 - \eta(\tau), & 0 \leq \tau \leq 1, \\ -m\Lambda_1 h(\Lambda_1(\tau - 1)), & 1 \leq \tau \leq \tau_b, \end{cases} \quad Z = \sin s z, \quad \alpha = s - \arcsin Z. \quad (7)$$

Extend $Z = \sin s$ before t_1 and $Z = 0$ after t_b . The two smooth pieces and extensions agree to every order, since the step is flat and h is supported in the interior. The source's selected pulse height in (3.12) is precisely $h_1 = m\Lambda_1 \sin s$; its allowed interval is $[0, c_p \Lambda_1 \sin s]$. The design gives

$$|Z| \leq \frac{1}{4}, \quad \zeta_{1,1} = Z, \quad \zeta_{1,2} = \sqrt{1 - Z^2} \geq \frac{\sqrt{15}}{4}, \quad \dot{\alpha} = -\frac{\dot{Z}}{\zeta_{1,2}}, \quad 0 \leq \alpha < \frac{1}{2}. \quad (8)$$

For the last bound, $Z \leq \sin s$ implies $\alpha \geq 0$, and $\arcsin(1/4) \leq (1/4)/\sqrt{1-1/16} < 1/3$ gives $\alpha \leq 1/8 + 1/3 < 1/2$. These are the source's actual feedback and the original laboratory phase component; gravity remains e_2 .

Define $\mathcal{P}$ first as the solution of the linear equation

$$\mathcal{P}_{\tau\tau} = z\mathcal{P}, \quad \mathcal{P}(0) = \mathcal{P}_\tau(0) = 1. \quad (9)$$

The same linear integral series proves existence, uniqueness, and smoothness for every trial on the full finite interval. Moreover the exact inverse map to the original amplitude pair is

$$\Theta_1(t) = -P_{\text{ent}}\mathcal{P}(\tau), \quad \Omega_1(t) = -\frac{K_1}{\nu}P_{\text{ent}}\mathcal{P}_\tau(\tau). \quad (10)$$

To check it, its component derivatives are exactly

$$-\Gamma P_{\text{ent}}\mathcal{P}_\tau = a_1\Omega_1, \quad -\frac{K_1}{\nu}\Gamma P_{\text{ent}}z\mathcal{P} = K_1 \sin s z\Theta_1 = b_1\Theta_1.$$

The data at t_1 match (6). Conversely the first amplitude equation with constant a_1 and $a_1b_1 = \Gamma^2z$ gives (9) for $-\Theta_1/P_{\text{ent}}$, with the stated initial derivative. Thus the auxiliary equation and original pair are explicitly inverse descriptions of the same dynamics, not a substituted oscillator.

On $0 \leq \tau \leq 1$, $z \geq 0$. Before any first zero of either $\mathcal{P}$ or $\mathcal{P}_\tau$, one has $\mathcal{P}_{\tau\tau} \geq 0$ and therefore $\mathcal{P}_\tau \geq 1$, $\mathcal{P} \geq 1 + \tau$. Continuity excludes that first zero and proves these inequalities through $\tau = 1$. Only now define $v = \mathcal{P}_\tau/\mathcal{P}$. It satisfies

$$v_\tau = 1 - \eta - v^2, \quad v(0) = 1.$$

Since $(v-1)_\tau = -(v+1)(v-1) - \eta$, the integrating-factor formula gives $v \leq 1$. Since $v > 0$ and $v_\tau \geq -v^2$, $(1/v)_\tau \leq 1$, so

$$\frac{1}{1+\tau} \leq v(\tau) \leq 1, \quad \log 2 \leq \int_0^1 v d\tau \leq 1, \quad v_d := v(1) \in [1/2, 1], \quad 2 \leq \mathcal{P}(1) \leq e. \quad (11)$$

On the negative pulse set $y = \Lambda_1(\tau - 1)$ and $\chi(y) = \mathcal{P}(1 + y/\Lambda_1)/\mathcal{P}(1)$. There is no division by an unknown evolving amplitude in this definition. Its exact equation and data are

$$\chi_{yy} + \frac{mh(y)}{\Lambda_1}\chi = 0, \quad \chi(0) = 1, \quad \chi_y(0) = v_d/\Lambda_1. \quad (12)$$

Before a possible first zero, χ is concave, so $\chi \leq 1 + y/\Lambda_1 \leq 2$. Since $mh/\Lambda_1 \leq c_p H/\Lambda_1 \leq 1/2$, one has $\chi_{yy} \geq -1$, hence

$$\chi(y) \geq 1 + \frac{v_d}{\Lambda_1}y - \frac{y^2}{2} \geq \frac{1}{2} \quad (0 \leq y \leq 1).$$

The strict positive lower bound excludes a first zero. Thus every trial satisfies on the entire pulse

$$\frac{1}{2} \leq \chi(y) \leq 1 + y/\Lambda_1, \quad \chi_y = \frac{v_d}{\Lambda_1} - \frac{m}{\Lambda_1} \int_0^y h(r)\chi(r) dr \geq -\frac{2c_p}{\Lambda_1}. \quad (13)$$

Consequently the original temperature is strictly negative for every trial, and the quotient is well defined. Writing $V(y; m) = v(1 + y/\Lambda_1; m) = \Lambda_1\chi_y/\chi$ gives

$$V_y = -mh(y) - \frac{V^2}{\Lambda_1}, \quad V(0; m) = v_d, \quad -4c_p \leq V(y; m) \leq v_d \leq 1. \quad (14)$$

Here is the parameter dependence needed to select the return. In the first-order integral system for (χ, χ_y) the coefficient matrix depends affinely on m . On the compact parameter interval, its iterated integrals converge uniformly with factorial denominators. Differentiating r times in m introduces at most a degree- r polynomial in the iteration index, still dominated by that factorial. Thus this series and each parameter derivative converge uniformly. The lower bound $\chi \geq 1/2$ then gives smooth parameter dependence of V . Differentiating its exact equation and using an integrating factor gives

$$\partial_m V(1; m) = - \int_0^1 h(y) \exp\left(-\frac{2}{\Lambda_1} \int_y^1 V(r; m) dr\right) dy < 0. \quad (15)$$

The strict sign follows because $h \geq 0$ has mass one and the exponential is positive. Indeed the explicit bounds also give $e^{-2/\Lambda_1} \leq -\partial_m V(1; m) \leq e^{8c_p/\Lambda_1}$. At $m = 0$, direct solution of $V_y = -V^2/\Lambda_1$ gives $V(1; 0) = v_d/(1 + v_d/\Lambda_1) > 0$. At $m = c_p$, integration gives $V(1; c_p) \leq v_d - c_p < 0$. Continuity and strict monotonicity therefore give exactly one $m_* \in (0, c_p)$ with $V(1; m_*) = 0$. This is exactly the source's selected return: by (10) and positivity, it is equivalent to $\Omega_1(t_b) = 0$.

For this selected trial V is nonincreasing and ends at zero, so $0 \leq V \leq v_d$. Integration of (14) now gives the exact identity and bounds

$$m_* = v_d - \frac{1}{\Lambda_1} \int_0^1 V(y; m_*)^2 dy, \quad \frac{1}{2} - \frac{1}{4\Lambda_1} \leq m_* \leq 1. \quad (16)$$

For the lower bound, $v_d - v_d^2/\Lambda_1$ is increasing for $v_d \in [1/2, 1]$, because its derivative is $1 - 2v_d/\Lambda_1 > 0$.

For every trial and every steering time,

$$0 \leq \log \mathcal{P}(\tau) \leq 1 + \Lambda_1^{-1}.$$

On the ramp this follows from (11); on the pulse it follows from $\mathcal{P}(1) \geq 2$, $\chi \geq 1/2$, and $\log \chi \leq \log(1 + \Lambda_1^{-1}) \leq \Lambda_1^{-1}$. For the selected trial the pulse logarithmic contribution is nonnegative, so the endpoint obeys

$$\log 2 \leq \log \mathcal{P}(\tau_b) \leq 1 + \Lambda_1^{-1}, \quad 2eP_* \leq P_b := P_1(t_b) \leq e^{2+\Lambda_1^{-1}} P_*. \quad (17)$$

The total duration from entry to return is exactly

$$t_b - t_0 = \frac{L + 2 + \Lambda_1^{-1}}{\nu \sin s}. \quad (18)$$

Throughout the selected stage P_1 is nondecreasing, $S \leq P_1 \leq P_b$, and $-(K_1/\nu)P_1 \leq \Omega_1 \leq 0$. For every unselected trial the valid bound instead is $|\Omega_1| \leq 4c_p(K_1/\nu)e^{2+\Lambda_1^{-1}}P_*$; its sign need not stay negative.

Near t_b , $h = 0$, so $Z = 0$ and $\alpha = s$. The equation $\mathcal{P}_{\tau\tau} = 0$, together with $\mathcal{P}_\tau(\tau_b) = 0$, implies that $\mathcal{P}$ is constant on that whole neighborhood. Thus α , Θ_1 , and Ω_1 match their stationary continuations to all orders. The endpoint state is

$$\alpha = s, \quad M_1 = R_s, \quad \zeta_1 = e_2, \quad w_1 = 1, \quad \Theta_1 = -P_b, \quad \Omega_1 = 0. \quad (19)$$

1.4 Control bounds and exact whole-stage integrals

For $j = 1, 2$ define the explicit finite profile quantities

$$B_j = \Gamma^j \sin s (\|\eta^{(j)}\|_\infty + c_p \Lambda_1^{j+1} \|h^{(j)}\|_\infty).$$

From (7), $|\partial_t^j Z| \leq B_j$. Differentiating the original control, including its denominator, gives

$$\dot{\alpha} = -\dot{Z}(1 - Z^2)^{-1/2}, \quad (20)$$

$$\ddot{\alpha} = -\ddot{Z}(1 - Z^2)^{-1/2} - Z\dot{Z}^2(1 - Z^2)^{-3/2},$$

$$|\dot{\alpha}| \leq \frac{4}{\sqrt{15}}B_1, \quad |\ddot{\alpha}| \leq \frac{4}{\sqrt{15}}B_2 + \frac{16}{15\sqrt{15}}B_1^2. \quad (21)$$

Higher derivatives exist because the displayed smooth composition has argument in $[-1/4, 1/4]$; no endpoint differentiability is omitted.

For exact coefficient costs use the invertible fixed map $Y = (\Theta_1, \nu\Omega_1/K_1)^T$. Its inverse is $(Y_1, K_1Y_2/\nu)^T$, and direct differentiation gives

$$\dot{Y} = \hat{B}Y, \quad \hat{B} = \begin{pmatrix} 0 & \Gamma \\ \nu Z & 0 \end{pmatrix}.$$

Let $J_\eta = \int_0^1 (1 - \eta(r)) dr \in [0, 1]$. On a hold of physical duration $T_h \geq 0$, the following are exact:

interval	duration	$\int \hat{B}_{12} dt$	$\int \hat{B}_{21} dt$
growth	L/Γ	L	L
transition	$1/\Gamma$	1	1
steering ramp	$1/\Gamma$	1	J_η
negative pulse	$1/(\Lambda_1\Gamma)$	$1/\Lambda_1$	m_*
holding	T_h	ΓT_h	0

For example, on the pulse substitution $y = \Lambda_1\Gamma(t - t_1 - \Gamma^{-1})$ gives $\int \nu|Z| dt = m_* \int_0^1 h = m_*$. Thus the two transition-and-steering costs are exactly $2 + \Lambda_1^{-1}$ and $1 + J_\eta + m_*$; their sum is at most $4 + c_p + \Lambda_1^{-1}$.

The activation and amplitude integrals needed by force estimates can also be retained without an asymptotic replacement. Set

$$J_a = \int_0^1 \eta'(r)e^r dr \in [1, e], \quad J_g = \int_0^1 \eta(r)e^r dr = e - J_a.$$

For the selected stage including the hold, direct change of variable and integration by parts give

$$\int_{t_0}^{t_b} |\dot{w}_1\Theta_1| dt = SJ_a, \quad \int_{t_0}^{t_b} |\dot{w}_1\Omega_1| dt = \frac{K_1}{\nu}SJ_a, \quad (22)$$

$$I_P := \int_{t_0}^{t_b+T_h} w_1 P_1 dt = \frac{S}{\Gamma}(J_g + e^{L+1} - e) + P_b T_h + \frac{eP_*}{\Gamma} \int_0^{1+\Lambda_1^{-1}} \mathcal{P}(\tau; m_*) d\tau, \quad (23)$$

$$I_\Omega := \int_{t_0}^{t_b+T_h} w_1 |\Omega_1| dt = \frac{P_b - SJ_a}{a_1}. \quad (24)$$

For the last identity, $P'_1 = -a_1\Omega_1 = a_1|\Omega_1|$, $w_1(t_0) = 0$, and $w_1(t_b + T_h) = 1$, hence $\int w_1 P'_1 = P_b - \int \dot{w}_1 P_1 = P_b - SJ_a$. In particular

$$I_P \leq \frac{S}{\Gamma}(J_g + e^{L+1} - e) + \frac{eP_*}{\Gamma}(1 + \Lambda_1^{-1})e^{1+\Lambda_1^{-1}} + P_b T_h.$$

These are finite specified integrals of the prescribed profiles and the uniquely selected exact linear solution.

1.5 The evolving background inherited by layer two

On the affine part of the first wave, the complete background for the next layer is

$$G_{<2} = -A_0 R_\alpha e_2 + w_1 K_1 \Theta_1 \zeta_1, \quad D_{<2} = \dot{\alpha} J + w_1 \Omega_1 J \zeta_1 \otimes \zeta_1. \quad (25)$$

These are source (3.3) with $A_1 = -K_1 \Theta_1 |\zeta_1|$ and the physical units retained. Their complete derivatives are

$$\dot{G}_{<2} = \dot{\alpha} J G_{<2} + (K_1 \dot{w}_1 \Theta_1 + w_1 A_0 \sin s \Omega_1) \zeta_1, \quad (26)$$

$$\begin{aligned} \dot{D}_{<2} = & \ddot{\alpha} J + (\dot{w}_1 \Omega_1 + w_1 K_1 \zeta_{1,1} \Theta_1) J \zeta_1 \otimes \zeta_1 \\ & + w_1 \Omega_1 \dot{\alpha} (-\zeta_1 \otimes \zeta_1 + J \zeta_1 \otimes J \zeta_1). \end{aligned} \quad (27)$$

Indeed $\dot{\zeta}_1 = \dot{\alpha} J \zeta_1$, $J \dot{\zeta}_1 = -\dot{\alpha} \zeta_1$, and the product rule gives each displayed term. In particular, using $J \zeta_1 \cdot G_{<2} = -A_0 \sin s$ gives the exact identity

$$\dot{G}_{<2} + D_{<2}^T G_{<2} = K_1 \dot{w}_1 \Theta_1 \zeta_1. \quad (28)$$

Thus the inherited gradient satisfies its true affine transport equation after activation, even while the first stage rotates it. The affine vorticity is $2\dot{\alpha} + w_1 \Omega_1$ and its scalar source is $G_{<2,1} = A_0 \sin \alpha + w_1 K_1 \Theta_1 \zeta_{1,1}$; therefore its exact vorticity residual is

$$(2\dot{\alpha} + w_1 \Omega_1)' - G_{<2,1} = 2\ddot{\alpha} - A_0 \sin \alpha + \dot{w}_1 \Omega_1. \quad (29)$$

At the selected endpoint define $A_1 = K_1 P_b > 0$. Then

$$G_{<2}(t_b) = (A_0 \sin s, -A_0 \cos s - A_1), \quad D_{<2}(t_b) = 0, \quad \sigma_1^2 = |G_{<2}(t_b)| = \sqrt{A_0^2 + A_1^2 + 2A_0 A_1 \cos s}. \quad (30)$$

Consequently

$$2eA_0 \lambda_1^{1/8} \leq A_1 \leq e^{2+\Lambda_1^{-1}} A_0 \lambda_1^{1/8}, \quad A_1 + A_0 \cos s \leq \sigma_1^2 \leq A_1 + A_0.$$

The lower norm bound is its positive vertical component in absolute value; the upper bound is the triangle inequality, which also follows by squaring the displayed expression. No horizontal component of the parent gradient has been discarded.

On the first hold the actual feedback is $F_1 = 0$. The endpoint state (19) is stationary under the original equations: $b_1 = K_1 \zeta_{1,1} = 0$ gives $\dot{\Omega}_1 = 0$, then $\dot{\Theta}_1 = a_1 \Omega_1 = 0$, and $\dot{\alpha} = 0$ gives $\dot{\zeta}_1 = 0$. Thus (30) remains the actual inherited background throughout this hold. For an arbitrary perturbation of the weighted amplitude pair its propagator over length T_h is exactly

$$\exp\left(T_h \begin{pmatrix} 0 & \Gamma \\ 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & \Gamma T_h \\ 0 & 1 \end{pmatrix}. \quad (31)$$

This follows by squaring the matrix and obtaining zero, and retains the nonzero first holding coefficient cost.

For clarity one can construct the next growth interval itself rather than prescribing an unspecified holding time. Retain $\lambda_2 = \lambda_1^{Q_2}$ with $Q_2 \geq 202$, an integer $k_2 \geq 0$, $L_2 = (k_2 + 41/8) \log \lambda_2$, and the source's

$$s_2 = L_2 \nu / \sigma_1, \quad K_2 = \mu \lambda_2, \quad \Gamma_2 = \sigma_1 \sin s_2.$$

An explicit sufficient finite choice ensuring $L_2 \geq 2$ and $0 < s_2 \leq 1/8$ is $\log \lambda_1 \geq 4096(k_2 + 41/8)Q_2$. To prove this, put $B = (k_2 + 41/8)Q_2$ and $x = \log \lambda_1$. The gain implies $\sigma_1 \geq \sqrt{2e\nu}e^{x/16}$; the

exponential series gives $e^{x/16} \geq x^2/512$. Hence $s_2 \leq 512B/(\sqrt{2}ex) \leq 1/8$. Also $L_2 = Bx \geq 4096B^2 > 2$.

During the second growth, $\alpha = s$, $\Omega_1 = 0$, and $D_{<2} = 0$, so $M_2 = I$, $\zeta_2 = e(s_2)$, and the original coefficient formulas yield exactly

$$a_2 = \frac{A_1 \sin s_2 + A_0 \sin(s + s_2)}{K_2}, \quad b_2 = K_2 \sin s_2, \quad \gamma_2 = \sqrt{[A_1 \sin s_2 + A_0 \sin(s + s_2)] \sin s_2} > 0. \quad (32)$$

Both signs are positive because $0 < s + s_2 \leq 1/4$. For its actual seed $S_2 = (A_0/\mu)\lambda_2^{-k_2-6}$ and entry vorticity $-\sqrt{b_2/a_2}S_2$, the exact solution is $P_2 = S_2 e^{\gamma_2(t-t_b)}$, $\Omega_2 = -\sqrt{b_2/a_2}P_2$. Substitution verifies both equations; the first threshold time is $t_{a,2} = t_b + L_2/\gamma_2$. Its activation is the source's $\eta(\Gamma_2(t - t_b))$. To see that activation completes before threshold, note

$$R := \frac{\gamma_2^2}{\Gamma_2^2} = \frac{A_1 + A_0 \cos s + A_0 \sin s \cot s_2}{\sigma_1^2}, \quad \cos s \leq R \leq 1 + \frac{\nu \sin s}{L_2 \sigma_1}. \quad (33)$$

For the lower bound, discard the positive cotangent term, use $\sigma_1^2 \leq A_1 + A_0$, and then $(A_1 + A_0 \cos s)/(A_1 + A_0) \geq \cos s$. For the upper bound use $A_1 + A_0 \cos s \leq \sigma_1^2$ and $\cot s_2 \leq 1/s_2$, the latter following from $(\sin r - r \cos r)' = r \sin r \geq 0$. Since $\sigma_1 \geq \nu$ and $L_2 \geq 2$, $R \leq 1 + \sin s/2 < 2$, and $\Gamma_2 L_2/\gamma_2 = L_2/\sqrt{R} > 1$. The exact first holding integral therefore is

$$H_1 = \Gamma \frac{L_2}{\gamma_2} = \sin s \frac{s_2}{\sin s_2} \frac{1}{\sqrt{R}}, \quad \frac{\sin s}{\sqrt{1 + \nu \sin s/(L_2 \sigma_1)}} \leq H_1 \leq \frac{\sin s}{\sqrt{\cos s}(1 - s_2^2/6)}. \quad (34)$$

Here $\sin r \leq r$ follows by integrating $\cos r \leq 1$; $\sin r \geq r - r^3/6$ follows by integrating $\cos r \geq 1 - r^2/2$, itself obtained from $1 - \cos r = \int_0^r \sin u \, du \leq r^2/2$. Thus no small-angle replacement is used in the bounds. The second transition also has $F_2 = 0$, because its only shear coefficient is $\Omega_1 = 0$; it prolongs the first pair's stationarity by exactly Γ_2^{-1} . The source counts that transition in the first layer's tail, not in H_1 .

1.6 Exact uncut forced fields and their retained diffusion cost

This subsection gives an explicit PDE realization of the just-proved finite stage on $\mathbb{R}^2$. Its fields have the spatial domain $\mathbb{R}^2$ and need not decay; compact spatial localization is a separate construction. Let F be any smooth odd 2π -periodic function. Define $Q_0(r) = \int_0^r F(v) \, dv$ and $\mathcal{Q}(r) = Q_0(r) - (2\pi)^{-1} \int_{-\pi}^{\pi} Q_0(v) \, dv$. Oddness gives zero period integral for F , so Q_0 is periodic; substitution $v = -u$ gives $Q_0(-r) = Q_0(r)$. Thus $\mathcal{Q}$ is even and mean-zero, and $\mathcal{Q}' = F$. Put $q(x, t) = K_1 \zeta_1(t) \cdot x$ and

$$\begin{aligned} \theta &= G_{<1} \cdot x + w_1 \Theta_1 F(q), \\ u &= \dot{\alpha} Jx + \frac{w_1 \Omega_1}{K_1} J \zeta_1 F(q), \end{aligned} \quad (35)$$

$$p = \frac{\dot{\alpha}^2}{2} |x|^2 + \frac{1}{2} x_2 (G_{<1} \cdot x) + \left(\frac{2w_1 \Omega_1 \dot{\alpha}}{K_1^2} + \frac{w_1 \Theta_1 \zeta_{1,2}}{K_1} \right) \mathcal{Q}(q). \quad (36)$$

An additive function of time can be subtracted if $p(0, t) = 0$ is desired. The exact inviscid forces are

$$f_\theta^0 = \dot{w}_1 \Theta_1 F(q), \quad f_u^0 = \left(\ddot{\alpha} - \frac{A_0}{2} \sin \alpha \right) Jx + \frac{\dot{w}_1 \Omega_1}{K_1} J \zeta_1 F(q). \quad (37)$$

We verify every component. The background derivative satisfies $\dot{G}_{<1} = \dot{\alpha} J G_{<1}$, hence its scalar transport residual vanishes. The phase satisfies $(\partial_t + \dot{\alpha} Jx \cdot \nabla)q = 0$. The wave velocity has scalar

product zero with ∇q , so it advects neither its own temperature nor its own velocity profile. Its advection of the background temperature equals $(w_1\Omega_1/K_1)(J\zeta_1 \cdot G_{<1})F(q)$, which cancels $w_1\dot{\Theta}_1 F(q)$ by (4). The remaining scalar residual is exactly $\dot{w}_1\Theta_1 F(q)$.

For momentum, write $C = w_1\Omega_1/K_1$ and $v = CJ\zeta_1 F(q)$. Differentiating v and adding its transport of $\dot{\alpha}Jx$ gives, after subtracting the wave buoyancy,

$$\left[\frac{\dot{w}_1\Omega_1}{K_1}J\zeta_1 + \frac{w_1\dot{\Omega}_1}{K_1}J\zeta_1 - 2C\dot{\alpha}\zeta_1 - w_1\Theta_1 e_2 \right] F(q).$$

The last three vector terms have zero scalar product with $J\zeta_1$: this is $w_1\dot{\Omega}_1/K_1 - w_1\Theta_1\zeta_{1,1} = 0$. Their scalar product with ζ_1 is $-2C\dot{\alpha} - w_1\Theta_1\zeta_{1,2}$. Since $(\zeta_1, J\zeta_1)$ is an orthonormal basis, the wave pressure in (36) cancels precisely these three terms, leaving the displayed activation force. The base momentum residual before pressure is $(\ddot{\alpha}J - \dot{\alpha}^2 I - e_2 \otimes G_{<1})x$. The gradient of the first two pressure terms removes its symmetric part. Its antisymmetric part is $(\ddot{\alpha} - G_{<1,1}/2)Jx = (\ddot{\alpha} - A_0 \sin \alpha/2)Jx$, as claimed. Finally $\text{div } u = 0$ since $\text{tr } J = 0$ and $J\zeta_1 \cdot \zeta_1 = 0$.

For the retained-viscosity equations

$$\partial_t \theta + u \cdot \nabla \theta = \kappa_{\text{phys}} \Delta \theta + f_\theta, \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \nu_{\text{phys}} \Delta u + \theta e_2 + f_u, \quad \nabla \cdot u = 0,$$

the very same state and controls solve them with the explicitly specified compensating forces

$$\begin{aligned} f_\theta &= f_\theta^0 - \kappa_{\text{phys}} w_1 \Theta_1 K_1^2 F''(q), \\ f_u &= f_u^0 - \nu_{\text{phys}} w_1 \Omega_1 K_1 J\zeta_1 F''(q). \end{aligned} \tag{38}$$

Indeed the affine Laplacians vanish, while two spatial derivatives of the profile multiply by $K_1^2 |\zeta_1|^2 = K_1^2$. This proves the asserted equations by substitution, without changing either ν or ν_{phys} and without assuming damping leaves the parent coefficients unchanged. The exact cumulative L_x^∞ norms of the two added forces through the entire stage and hold are

$$\kappa_{\text{phys}} K_1^2 \|F''\|_\infty I_P, \quad \nu_{\text{phys}} K_1 \|F''\|_\infty I_\Omega. \tag{39}$$

Equality holds because $x \mapsto K_1 \zeta_1 \cdot x$ is surjective onto $\mathbb{R}$ and $|J\zeta_1| = 1$; this verifies the norm statement as well as an upper bound. The scalar holding contribution is exactly $\kappa_{\text{phys}} K_1^2 \|F''\|_\infty P_b T_h$; the velocity holding contribution is zero. These finite costs are not asserted to be summable over the source's infinite frequency sequence.

The construction proves a finite controlled stage, its actual evolving affine background, its return, and its next growth holding interval. It does not establish an infinite viscous cascade or a smooth terminal-time viscous force. In particular, simply replacing the first temperature by an exponentially damped amplitude during holding would replace A_1 in (30) by a time-dependent quantity and would change both coefficients and the rate in (32). The compensating forces in (38) give an exact specified relation that retains the original coupled finite trajectory; their costs have been retained explicitly rather than claimed negligible.

2 Solving the next growth interval with a diffusing parent

This section continues an uncompensated equal-diffusion variant of the first stage through the next growth interval. It keeps the laboratory direction and the original coefficients. The exact amplitude system is realized below both on a forced affine parent and on an actual two-wave state with an explicitly computed interaction force. The latter force specifies the difference between the affine-gradient calculation and the full spatial equations.

2.1 The inherited gradient and the original second pair

Retain the preceding section's $A_0, \mu, \nu, K_1, s, \lambda_1$, and let $\nu_{\text{phys}} = \kappa_{\text{phys}} = d > 0$. For a sine first wave on its rotating affine base, the exact amplitude equations with retained diffusion are

$$\dot{\Theta}_{1,d} = a_1 \Omega_{1,d} - dK_1^2 \Theta_{1,d}, \quad \dot{\Omega}_{1,d} = b_1 \Theta_{1,d} - dK_1^2 \Omega_{1,d}.$$

If (Θ_1, Ω_1) denotes the first inviscid pair of the preceding section on its specified control and times, direct differentiation proves

$$(\Theta_{1,d}, \Omega_{1,d})(t) = e^{-dK_1^2(t-t_0)}(\Theta_1, \Omega_1)(t).$$

Both the forward map and its inverse multiplication by the positive exponential satisfy the indicated systems, so they are a bijection of solution spaces. In particular the selected vorticity still returns to zero at the specified t_b . This uses the inviscid times and does not assert the damped temperature crosses the same growth threshold. Its exact log-gain from the original seed is

$$\log(P_b/S) - \frac{dK_1^2}{\nu \sin s}(L + 2 + \Lambda_1^{-1}).$$

Set $P_{d,b} = e^{-dK_1^2(t_b-t_0)}P_b > 0$, $A_{1,d} = K_1 P_{d,b}$, and $r = t - t_b \geq 0$. During the fixed first holding control $\alpha = s$ the exact first pair is

$$\Theta_{1,d}(r) = -P_{d,b}e^{-qr}, \quad \Omega_{1,d} = 0, \quad q = dK_1^2, \quad G_d(r) = (A_0 \sin s, -A_0 \cos s - A_{1,d}e^{-qr}).$$

The gradient therefore changes at the rate $G'_d(r) = (0, qA_{1,d}e^{-qr})$. Let the new laboratory phase be $\zeta_2 = e(s_2)$ with $0 < s_2 \leq 1/8$, and let $K_2 = \mu\lambda_2 > 0$. The actual first hold has zero velocity gradient, so $\zeta'_2 = 0$. The second affine-wave coefficients are exactly

$$a_{2,d}(r) = -\frac{J\zeta_2 \cdot G_d(r)}{K_2} = \frac{A_{1,d}e^{-qr} \sin s_2 + A_0 \sin(s + s_2)}{K_2}, \quad b_2 = K_2 \sin s_2, \quad \delta_2 = dK_2^2. \quad (40)$$

Their signs are positive because $0 < s + s_2 \leq 1/4$. The exact second sine amplitude system is

$$\Theta'_2 = a_{2,d}(r)\Omega_2 - \delta_2\Theta_2, \quad \Omega'_2 = b_2\Theta_2 - \delta_2\Omega_2. \quad (41)$$

In particular the changing inherited gradient has not been replaced by its endpoint value. An affine realization alone would require the scalar force $G'_d(r) \cdot x$: the affine Laplacian is zero, so that force cannot be inferred to vanish from the old sine-wave diffusion.

2.2 Exact operator change and convergent fundamental solutions

Define

$$c_0 = A_0 \sin(s + s_2) \sin s_2 > 0, \quad c_1 = A_{1,d} \sin^2 s_2 > 0, \quad W(r) = e^{\delta_2 r} \Omega_2(r).$$

Because b_2 is constant, differentiating the second original equation gives the exact system equivalence

$$W'' = (c_0 + c_1 e^{-qr})W, \quad \Omega_2 = e^{-\delta_2 r} W, \quad \Theta_2 = \frac{e^{-\delta_2 r}}{b_2} W'. \quad (42)$$

Conversely these last two formulas give $\Omega'_2 = b_2\Theta_2 - \delta_2\Omega_2$ and $\Theta'_2 = e^{-\delta_2 r} W''/b_2 - \delta_2\Theta_2 = a_{2,d}\Omega_2 - \delta_2\Theta_2$, since $b_2 a_{2,d} = c_0 + c_1 e^{-qr}$. Thus the initial-data map is the invertible map $(\Theta_2(0), \Omega_2(0)) \mapsto (W(0), W'(0)) = (\Omega_2(0), b_2\Theta_2(0))$.

Put

$$x(r) = x_0 e^{-qr/2}, \quad x_0 = \frac{2\sqrt{c_1}}{q}, \quad \vartheta = \frac{2\sqrt{c_0}}{q} > 0, \quad W(r) = \mathcal{W}(x(r)).$$

Here $x(r)$ is a scalar change of independent variable and is unrelated to either physical spatial coordinate x_1, x_2 , which remain fixed in the PDE below. Since $x' = -qx/2$ and $x'' = q^2 x/4$, one has

$$W'' - (c_0 + c_1 e^{-qr})W = \frac{q^2}{4} [x^2 \mathcal{W}_{xx} + x \mathcal{W}_x - (x^2 + \vartheta^2) \mathcal{W}].$$

This proves the operator transformation including its nonzero factor. Rather than invoking a special-function name, define on $x > 0$

$$f(x) = x^\vartheta \sum_{n=0}^{\infty} a_n x^{2n}, \quad a_0 = 1, \quad a_n = \frac{a_{n-1}}{4n(n+\vartheta)}, \quad g(x) = f(x) \int_{x_0}^x \frac{d\xi}{\xi f(\xi)^2}. \quad (43)$$

For every compact interval in $x > 0$, the consecutive-term absolute ratio $x^2/[4n(n+\vartheta)]$ tends uniformly to zero. Every fixed derivative multiplies these terms by a polynomial in n and a bounded power of x^{-1} on that compact interval; its series therefore converges uniformly as well. Termwise substitution is valid. The coefficient of $x^{\vartheta+2n}$ in $x^2 f'' + x f' - (x^2 + \vartheta^2) f$ is $4n(n+\vartheta)a_n - a_{n-1} = 0$ for $n \geq 1$; the $n = 0$ coefficient is also zero. All coefficients are positive, so $f > 0$. The defining integral for g consequently exists on every compact subinterval of $x > 0$. If $I' = 1/(xf^2)$, differentiation yields

$$f(fI)' - f'(fI) = f^2 I' = 1/x.$$

Also $I''/I' = -1/x - 2f'/f$; substituting $g = fI$ into $g'' + g'/x - (1 + \vartheta^2/x^2)g$ leaves $fI'' + (2f' + f/x)I' = 0$. Thus g solves the same equation and its Wronskian with f is the nonvanishing function $1/x$. For completeness this proves completeness of the two solutions: at any $x > 0$ their value-and-derivative matrix is invertible, so one linear combination matches arbitrary Cauchy data; linear uniqueness then identifies that combination with any solution on overlapping compact subintervals of $(0, \infty)$.

For the original data let $W_0 = \Omega_2(0)$ and $W_1 = b_2 \Theta_2(0)$. The exact solution is

$$W(r) = c_f f(x(r)) + c_g g(x(r)), \quad c_f = \frac{W_0}{f(x_0)}, \quad c_g = -\frac{2f(x_0)}{q} \left(W_1 + \frac{qx_0}{2} c_f f'(x_0) \right). \quad (44)$$

Indeed $g(x_0) = 0$, $g'(x_0) = 1/[x_0 f(x_0)]$, and $x'(0) = -qx_0/2$ verify both initial conditions directly. For every finite r , $x(r) > 0$, so the displayed functions and their derivatives are finite there. If $d = 0$, the preceding substitutions dividing by q are not used: the original equation is $W'' = (c_0 + c_1)W$, with the exact solution $W = W_0 \cosh(\sqrt{c_0 + c_1} r) + W_1 \sinh(\sqrt{c_0 + c_1} r)/\sqrt{c_0 + c_1}$.

2.3 Signs and quantitative physical gain with this feedback

Let $\gamma_0 = \sqrt{c_0}$ and $\gamma_i = \sqrt{c_0 + c_1}$. Use the original instantaneous growing eigenline at entry: $\Theta_2(0) < 0$ and $\Omega_2(0) = \sqrt{b_2/a_{2,d}(0)} \Theta_2(0) < 0$. Then $Y = -W$ has $Y(0) = Y_0 > 0$ and $Y'(0) = \gamma_i Y_0 > 0$. As long as $Y > 0$, its equation gives $Y'' > 0$ and $Y' \geq Y'(0) > 0$; this excludes a first zero by continuity. Define

$$Y_- = Y_0 \left(\cosh(\gamma_0 r) + \frac{\gamma_i}{\gamma_0} \sinh(\gamma_0 r) \right), \quad Y_+ = Y_0 e^{\gamma_i r}.$$

The differences have zero initial value and derivative and satisfy

$$(Y - Y_-)'' - \gamma_0^2 (Y - Y_-) = c_1 e^{-qr} Y,$$

$$(Y_+ - Y)'' - \gamma_i^2(Y_+ - Y) = c_1(1 - e^{-qr})Y.$$

For $a > 0$, the formula $Z(r) = \int_0^r \sinh(a(r-u))S(u)/a \, du$ has $Z(0) = Z'(0) = 0$ and $Z'' - a^2Z = S$, by two differentiations. Uniqueness proves it is the solution of that initial-value problem. Both sources above are nonnegative; both this kernel and its first r derivative are nonnegative for $0 \leq u \leq r$. Therefore $Y_- \leq Y \leq Y_+$ and $Y'_- \leq Y' \leq Y'_+$ for every finite $r \geq 0$. The inverse physical map now gives both amplitudes negative and the precise temperature-gain bounds

$$e^{-\delta_2 r} \left(\cosh(\gamma_0 r) + \frac{\gamma_0}{\gamma_i} \sinh(\gamma_0 r) \right) \leq \frac{-\Theta_2(r)}{-\Theta_2(0)} \leq e^{(\gamma_i - \delta_2)r}. \quad (45)$$

In particular, if the actual numerical coefficients satisfy $\delta_2 \geq \gamma_i$, this particular growth control never raises the second temperature above its entry amplitude. This is a proved obstruction to that specified uncompensated control, not a statement about other controls or the existence of an infinite viscous sequence.

2.4 The exact map to spatial fields and its interaction force

The affine parent $\theta_b = G_d(r) \cdot x$, $u_b = 0$ is a solution with pressure $p_b = x_2 G_d(r) \cdot x/2$ and forces $f_{\theta,b} = G'_d(r) \cdot x$ and $f_{u,b} = -(A_0 \sin s/2)Jx$. The scalar statement follows from $\Delta\theta_b = 0$. For momentum, direct differentiation gives $\nabla p_b - \theta_b e_2 = (G_d x_2 - e_2(G_d \cdot x))/2 = -G_{d,1}Jx/2$. An added sine wave whose amplitudes solve (41) has zero additional PDE residual except for its activation, by the same explicit phase, transverse-vector, and pressure calculations below. Thus the solved system has an exact forced affine-background realization.

There is also a full two-wave realization specifying the original spatial discrepancy. Let F_1, F_2 be smooth odd periodic profiles, each with mean-zero primitive $Q'_j = F_j$, and assume $F'_1(0) = 1$. This includes both sine and the released profile which is linear near zero. Set

$$G_0 = (A_0 \sin s, -A_0 \cos s), \quad T(r) = -P_{d,b}e^{-qr}, \quad q_1 = K_1 x_2, \quad q_2 = K_2 \zeta_2 \cdot x,$$

and let $w_2(r)$ be any specified smooth activation, for example the source's $\eta(\Gamma_2 r)$. Define on $\mathbb{R}^2$

$$\begin{aligned} \theta &= G_0 \cdot x + T F_1(q_1) + w_2 \Theta_2 F_2(q_2), \\ u &= \frac{w_2 \Omega_2}{K_2} J \zeta_2 F_2(q_2), \\ p &= \frac{1}{2} x_2 (G_0 \cdot x) + \frac{T}{K_1} Q_1(q_1) + \frac{w_2 \Theta_2 \zeta_{2,2}}{K_2} Q_2(q_2). \end{aligned} \quad (46)$$

Their complete retained-diffusion forces are

$$\begin{aligned} f_\theta &= -dK_1^2 T [F_1(q_1) + F_1''(q_1)] + w'_2 \Theta_2 F_2(q_2) \\ &\quad - dK_2^2 w_2 \Theta_2 [F_2(q_2) + F_2''(q_2)] + f_{\text{cross}}, \end{aligned} \quad (47)$$

$$f_{\text{cross}} = \frac{w_2 \Omega_2}{K_2} \sin s_2 K_1 T [F'_1(q_1) - 1] F_2(q_2), \quad (48)$$

$$f_u = -\frac{A_0 \sin s}{2} Jx + \frac{w'_2 \Omega_2}{K_2} J \zeta_2 F_2(q_2) - dK_2 w_2 \Omega_2 J \zeta_2 [F_2(q_2) + F_2''(q_2)]. \quad (49)$$

Here is a direct verification. The old wave has derivative $T' = -dK_1^2 T$ and Laplacian $K_1^2 T F_1''$, giving the first scalar force. The old temperature gradient differs from the coefficient gradient $G_d = G_0 + K_1 T e_2$ by precisely $K_1 T [F'_1(q_1) - 1] e_2$. Its advection by u is the displayed f_{cross}

because $J\zeta_2 \cdot e_2 = \sin s_2$. The remainder of that old-temperature advection is $(w_2\Omega_2/K_2)J\zeta_2 \cdot G_d F_2 = -w_2 a_{2,d}\Omega_2 F_2$; it cancels the coupling part of $w_2\Theta'_2 F_2$. The new wave advects its own temperature and velocity by zero since $J\zeta_2 \cdot \zeta_2 = 0$. Its amplitude damping and its Laplacian give the two new-profile terms in the forces. The old wave pressure has gradient $TF_1(q_1)e_2$, so it cancels old wave buoyancy exactly. The new pressure has gradient $w_2\Theta_2\zeta_{2,2}\zeta_2 F_2$. The new momentum coupling vector before that pressure is $w_2\Theta_2(\zeta_{2,1}J\zeta_2 - e_2)F_2 = -w_2\Theta_2\zeta_{2,2}\zeta_2 F_2$, by resolving e_2 in the orthonormal basis $(\zeta_2, J\zeta_2)$. Hence it cancels as well. The base pressure supplies exactly the first vector-force term, as proved above, and divergence is zero by transversality. This verifies both full equations including all interactions.

For the sine profiles, $F_j + F_j'' = 0$ identically. The remaining scalar interaction force is then exactly

$$f_{\text{cross}} = -\frac{A_{1,d}e^{-qr}}{K_2}w_2\Omega_2 \sin s_2 [\cos(K_1 x_2) - 1] \sin(q_2), \quad \|f_{\text{cross}}(r)\|_\infty \leq \frac{2A_{1,d}e^{-qr}}{K_2}|w_2\Omega_2| \sin s_2.$$

The bound follows from $|\cos u - 1| \leq 2$ and $|\sin v| \leq 1$. For the released linear-core profile the same explicit formulas hold, and $F'_1 - 1 = 0$ on its original linear interval, so the interaction force vanishes there. Its old-profile diffusion defect is instead $-dK_1^2 T q_1$ on that interval, since $F_1(q_1) = q_1$ and $F_1''(q_1) = 0$. Its gradient is exactly $G'_d(r)$. Thus exponential damping of that released linear core requires the stated core force, and is not an unforced consequence of its zero Laplacian. The explicit forces, inverse solution maps, and convergent formulas above carry the feedback calculation to an actual spatial solution without conflating the sine state, its affine gradient, and the released profile. No compact-support or infinite-iteration claim is made in this section.